

very expensive. Movies had to be shot with two cameras. In general, a lot of time, money and effort was invested in the production, broadcast and viewing of 3D video content. Yet, that did not dampen the 3D fantasy one bit, for the makers or for the viewers!

Today, the consumer electronics industry seems keen on killing 3D's novelty value and making it a living room technology instead. Apparently, the 3D hype reached a feverish pitch last year, with foreign markets flooded with products small and big—3D televisions, 3D monitors, 3D video games, 3D movies, 3D glasses, and even 3D goggles that can add a third dimension to games you play on your iPhone or iPod. 3D televisions (3D TVs) seem to be leading this pack of products.

It is good for us that the effect has not engulfed the Indian market at such a rapid pace, as we can wait and watch how well the technology matures before pulling out our credit cards. Indeed, experts are wary of the speed at which 3D is being canvassed, since the technology and the price points have a long way to go.

Nevertheless, we are not going to be spared the craze, especially on the television front. Almost every brand present in India is talking about its 3D plans, some having launched basic models already.

“3D technology reproduces on screen action as we see in real life, making the images more real and hence allowing far greater viewer engagement and enjoyment. 3D is considered the next technology evolution after colour replaced B&W, and high-definition (HD) came to 2D,” comments Manish Sharma, director-marketing, Panasonic India.

While today's 3D television models continue to require special glasses for 3D viewing, these are definitely far more capable and less cumbersome than the ones of yesteryears. Auto-stereoscopic models, which do not require glasses, are also coming out. Light-emitting diode (LED) and plasma models of 3D TVs, special cameras

How is 3D content broadcast?

Frame-compatible plano-stereoscopic technique. This technique—used by Sky, Orange, ESPN and DirecTV—helps operators get out and about 3D quickly. Here, the left and right HD images are multiplexed (either side-by-side, top-and-bottom or otherwise) into a single HD signal, which is decoded back into 3D by the viewer's TV set. The user needs to have a 3D-ready TV, but no special set-top box.

While this seems advantageous, there are two hidden disadvantages. First, since the two HD signals are compressed into one, the resultant 3D image loses up to half

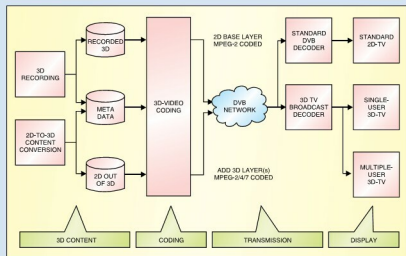


Fig. 1: 3D broadcasting scheme; Image courtesy: Wikipedia

of its original resolution. Second, the broadcast version can be viewed properly only on 3D TVs and is useless when seen on a 2D set. Hence the operator would have to broadcast two versions (3D and 2D) of the same channel, necessitating two expensive, bandwidth-intensive satellite transponders.

Service-compatible format. This is probably the technique that will work in the long run. Here, the left-eye signal is broadcast as a standard MPEG picture, while the data for the second eye is derived using the left as reference (this is also called 2D+Delta or 2D+Difference). Such a system would require new decoders to be built into the set-top box. This can also be seen normally on a conventional 2D set, without the 3D experience, relaxing the pressure on operators to broadcast two versions of the same channel.

While the requirements for frame-compatible stereo TV have been agreed upon by the DVB Project, the consideration of specifications for service-compatible modes is still underway. The frame-compatible model is also easier for broadcasters to begin with.

for easy shooting of 3D content, real-time 2D-to-3D converters and more such technologies are making an entry.

Let's take a quick look at the basics behind 3D televisions and some of the latest technologies in this space, to help you prepare for all the brochures that will soon flood the markets.

Stereoscopy: Inspired by the eye

Our world is in 3D; our eyes see in 3D. And, that is precisely what makes the

'3D' experience immersive and closer to real-life than the best of 2D photos, videos or games.

Most of the products that are being launched today are based on techniques of stereoscopy, inspired largely by the working of our eye. Our eyes are like two lenses, each capturing the same image from different angles. The brain processes and merges the two images, creating a 3D image for us. This processing happens so fast that we do not even realise it happening.